

Testing and Validation

Test Cases Summary

Test Case ID	Scenario	Input	Expected Result	Status
TC-01	Generate SQL Query	Show total sales by month	Valid SQL query generated and executed successfully	Pass
TC-02	Product Revenue Analysis	Which product has highest sales?	Product-wise revenue displayed correctly	Pass
TC-03	Employee Analytics	Show employee count by department	Department-wise employee count returned	Pass
TC-04	Monthly Trend Analysis	Monthly sales trend in 2024	Line chart generated successfully	Pass
TC-05	CSV Export	Export query results	CSV file downloaded successfully	Pass
TC-06	Security Validation	DROP TABLE employees	Query blocked and warning displayed	Pass
TC-07	Security Validation	DELETE FROM sales	Query blocked and warning displayed	Pass
TC-08	Invalid Query Handling	Query non-existent table	Appropriate error message displayed	Pass

Security Testing

Test Case	Input	Expected Result	Status
SQL Injection Attempt	SELECT * FROM employees; DROP TABLE employees;	Blocked	Pass
Multiple Statements	SELECT * FROM sales; DELETE FROM sales;	Blocked	Pass
Unauthorized Modification	UPDATE products SET price = 0	Blocked	Pass
Unauthorized Deletion	DELETE FROM customers	Blocked	Pass

Result Summary

- All functional test cases were executed successfully.
- Natural language queries were correctly converted into SQL queries.
- Data visualization components generated accurate charts and graphs.
- CSV export functionality worked as expected.
- Security mechanisms successfully prevented destructive SQL operations.
- Invalid queries were handled gracefully with meaningful error messages.

Conclusion

The AnalyticsGPT Agent successfully met all functional and security requirements. The system accurately generated SQL queries from natural language inputs, provided analytical insights through visualizations, enabled secure data export, and effectively protected the database against unauthorized or malicious operations.